Assignment: From Raw Records to Analysis-Ready Data

Hospital Readmission Data: an end-to-end data preprocessing pipeline

1. Scenario and goal

You are a data analyst in the quality office of a hospital group. Management wants to know which patient, stay, laboratory and medication factors are associated with **readmission within 30 days of discharge**. Four systems (registration, hospital information system, laboratory, pharmacy) each produced their own extract. Nobody has checked them, and they were never designed to fit together.

Your job is **not** to model readmission. Your job is to turn the four raw extracts into one trustworthy, documented, analysis-ready dataset, and to **prove with evidence** that every preprocessing decision improved the data, or at least did not damage it.

2. The required pipeline

Raw extracts → **Stage 1** Integration → **Stage 2** Cleaning → **Stage 3** Transformation → **Stage 4** Discretization → Reduction

3. Part 1: Data Integration (20 points)

Task 1.1 Source profiling and schema matching (4 pts). Profile every source (rows, columns, types as found, distinct values, blanks, sentinel-like values). Produce a schema-matching table: for every attribute that appears in more than one source, or that links sources, list the source columns, the differences in name, format and unit, and the documented versus actually observed content. List at least five discrepancies between the data dictionary and the data.

Task 1.2 Keys and referential integrity (4 pts). Identify all linking keys. Report join coverage (share of rows that find a partner) for each link *before* any repair, then repair the keys and report coverage *after*. Explain what a naive join would have done to your data. Decide and justify how to treat rows that still have no partner.

Task 1.3 Entity resolution for patients (6 pts). The same person may appear in `patients.csv` more than once.

- Design blocking (to avoid comparing all pairs) and a similarity score that combines at least name similarity, date-of-birth comparison, and one further attribute.
- Choose thresholds for 'match', 'review' and 'non-match' using evidence (a score histogram).
- Estimate precision by manually labelling a random sample of 50 predicted pairs, and estimate missed matches by manually inspecting 50 near-threshold pairs. Report both honestly.
- Assign a resolved `person_id` to every patient record and every admission.
- Define survivorship rules for conflicting attribute values (table: attribute, rule, rationale) and apply them.
- Then create the split (rule 5): about 70/30 by `person_id`, with a fixed seed.

Task 1.4 Integrating labs and medications (3 pts). Bring both sources to the grain of one row per admission. Harmonise test names; choose an aggregation rule for repeated measurements (first, last, mean, max, number of draws) separately for each test and defend it; flatten the nested prescription records; and describe what you do with admissions that have no lab or pharmacy record (a missing record is not the same as a normal result).

Task 1.5 Redundancy and cross-source conflicts (3 pts). Find attributes that carry the same information, within or across sources. For each redundant pair or cross-source disagreement, state which version you keep or how you reconcile them, and log the number of conflicts.

Deliverable of Part 1: `stage1_integrated` (one row per admission), the conflict log, the schema-matching table.

Reflection questions: Why might a wrong merge be worse than a missed merge here? What would change if you resolved entities after computing prior-visit counts?

4. Part 2: Data Cleaning (25 points)

Task 2.1 Syntactic standardisation (4 pts). Normalise labels (case, whitespace, spelling variants, abbreviations), data types, sentinel codes (convert every missing-value encoding you find to a true missing value and list each one), date formats (document how you resolved ambiguous formats and verify with a second source) and units (convert every measurement to one documented unit; detect values whose unit label does not match their magnitude). Show before/after value counts or distributions for at least six columns.

Task 2.2 Duplicates (3 pts). Identify all patterns of duplicate records in admissions and labs. Define a resolution rule for each, and justify when you keep the first, keep the latest, merge, or drop all.

Task 2.3 Invalid and inconsistent values (4 pts). Write at least ten explicit validation rules covering chronology, plausible ranges, cross-field consistency and domain constraints. Think about the cohort too: is every stay eligible for 30-day readmission? Produce a violation table (rule, rows violated, share) and decide per rule: repair (and how), flag, set to missing, or remove.

Task 2.4 Missing data: diagnosis (4 pts). For every column with missing values, report the rate; whether the missingness is structural (the value cannot exist), recoverable from another source, or genuinely missing; and your evidence for MCAR, MAR or MNAR (for example compare other variables between rows with and without a value using t-tests, Mann–Whitney or chi-square; a missingness heatmap; Little's MCAR test is optional). Explain why MNAR cannot be proven from the observed data alone and what you can argue instead.

Task 2.5 Missing data: treatment (6 pts). Choose two numeric variables (one roughly symmetric, one skewed) and one categorical variable with substantial missingness. Find a method and fill them.

Task 2.6 Outliers (4 pts). On at least five variables (at least one from patients, three from admissions and one from labs), detect outliers. Show summary statistics and skewness before and after. Discuss one case where removing outliers would have been harmful.

Deliverable of Part 2: `stage2_clean`, validation-rule table, missingness report, outlier decision table.

Reflection question: Which of your cleaning choices could change the readmission rate you report? Quantify one.

5. Part 3: Data Transformation (20 points)

Task 3.1 Feature construction (5 pts). Construct at least ten new features covering: demographics (for example age at admission, BMI), utilisation history built from the *resolved* patient record, clinical burden from diagnoses, aggregated laboratory results, medication information, and stay characteristics. For each feature document the definition, source columns, expected direction of association, and a one-line **leakage statement** (could this value be known at discharge?). History features must be *point-in-time correct*: they may only use admissions that finished before the index admission started. No feature may use the target of the index admission or of any later one.

Task 3.2 Skewness and distribution shape (4 pts). For at least six numeric variables, measure skewness and inspect histograms and Q–Q plots. Apply at least three transformations (for example log1p, square root, Box–Cox, Yeo–Johnson), tabulate skewness before and after, and choose per variable, including variables you deliberately leave untransformed and why (zeros, bounded values, counts, already symmetric). Estimate any lambda from the training set only.

Task 3.3 Scaling (3 pts). Apply min–max, z-score and robust scaling to at least six numeric features. (a) Show how each reacts to the outliers you decided to keep (range compression, centre shift). (b) Take 2,000 random pairs of training rows, compute Euclidean distances on unscaled data and on each scaled version, and report which features dominate the squared distance in each case. Choose a scaler per feature group, and state which later techniques need scaling and which do not.

Task 3.4 Encoding categorical variables (5 pts). Classify every categorical or identifier-like column (nominal, ordinal, numeric-coded but categorical, high-cardinality) and justify the encoding. Implement yourself: one-hot encoding with rare-level grouping and a stated reference level; frequency encoding; and **smoothed target (mean) encoding** (m-estimate). Then run the **leakage experiment**: encode the primary diagnosis code (a) with the encoding fitted on all rows and evaluated on the same rows, and (b) fitted on the training set and evaluated on the test set. Compare how strongly each encoded column correlates with the target, and explain the gap.

Task 3.5 Nested and text data (3 pts). Normalise drug names (generic, brand, strength, typos) to a single generic name using a dictionary plus fuzzy matching, map generics to drug classes, and build multi-hot or count features. Normalise the multi-valued allergy field. Audit the free-text discharge notes: list every kind of statement that refers to events *after* discharge and show how you removed or excluded them.

Deliverable of Part 3: stage3_transformed, feature dictionary, transformation decisions table, the leakage experiment figure.

6. Part 4: Data Discretization (15 points)

Work with at least four numeric variables (including age, one utilisation count and one laboratory aggregate).

Task 4.1 Unsupervised binning (4 pts). Implement equal-width and equal-frequency binning yourself, and cluster-based binning with one-dimensional k-means (library allowed), for k = 3, 5 and 8. Report bin edges, bin sizes, empty or tiny bins, and how outliers and ties affect each method.

Task 4.2 Binning (4 pts). Use entropy-based discretization.

Task 4.3 Concept hierarchies (3 pts). Build a hierarchy for diagnosis codes (full code → 3-digit category → chapter, plus your own clinical grouping if useful) and for age (clinically meaningful groups). For each level report the number of distinct values, the share of rows in values with fewer than 30 records, and the information gain with the target (train and test). Propose an adaptive roll-up (for example replace a code by its parent when support falls below a threshold) and justify the threshold.

Deliverable of Part 4: stage4_discretized, binning comparison table, hierarchy tables.

7. Deliverables

Item	Content
report.pdf (6–8 pages plus appendices)	Summary; pipeline overview with the row waterfall; decisions per stage; key figures; leakage audit; findings, limitations; decisions table (Appendix B)
Code/Model	Narrative analysis with all experiments, executed top to bottom
outputs/	stage1_integrated, stage2_clean, stage3_transformed, stage4_discretized

11. Grading

Part	Points
1 Integration	25
2 Cleaning	30
3 Transformation	25

Part	Points
4 Discretization	20
Total	100

Appendix A: Dataset description

Context. The data come from a fictitious hospital group and cover roughly 2018–2024. All patients are synthetic, but the structure imitates real hospital extracts. Four independent systems each exported their own file. The files were never harmonised, reconciled or quality-checked, and no ground truth is supplied. The descriptions below state the *intended* content of each field, and the delivered data may deviate from it.

A.1 Files at a glance

File	Source system	One row / record is	Approx. size	Format
patients.csv	Registration	A patient registration record	about 12,000 rows, 10 columns	CSV
admissions.csv	Hospital information system	A hospital admission record	about 20,000 rows, 28 columns	CSV
labs.csv	Laboratory	A single laboratory result	about 190,000 rows, 6 columns	CSV
medications.json	Pharmacy	The pharmacy record of one admission	about 19,000 records	JSON (nested)
data_dictionary.md		Short field documentation		Markdown

Sizes vary slightly between students.

A.2 How the files relate

- patients.patient_id ↔ admissions.pat_id (one person, many admissions).
- admissions.admission_id ↔ labs.admission_id (one admission, many lab results).
- admissions.admission_id ↔ medications.json field admission_id (one admission, one pharmacy record).
- Records are not guaranteed to follow one uniform layout in every file.

A.3 patients.csv

Column	Description
patient_id	Registration record ID ('P' followed by 6 digits)
first_name, last_name	Name as registered
sex	Sex recorded at registration
date_of_birth	Date of birth
race	Self-reported race / ethnicity
weight, weight_unit	Weight at registration and its unit

Column	Description
height	Height in cm
postal_code	5-digit ZIP code of residence

A.4 admissions.csv

Column	Description
admission_id	Admission identifier
pat_id	Patient identifier (links to patients.patient_id)
admit_date, discharge_date	Dates of admission and discharge
age	Age in years at admission
gender	Patient gender
admission_type_id, discharge_disposition_id, admission_source_id, ward_code	Coded administrative fields (code lists in A.7)
payer_code	Insurance payer
medical_specialty	Specialty of the admitting physician
time_in_hospital	Length of stay in days
los_hours	Length of stay in hours
num_lab_procedures, num_procedures, num_medications	Counts of laboratory tests, procedures and medications during the stay
number_outpatient, number_emergency, number_inpatient	Outpatient, emergency and inpatient visits in the 12 months before admission
diag_1, diag_2, diag_3	Primary, secondary and tertiary diagnosis (ICD-9-CM code)
number_diagnoses	Number of diagnoses entered for the stay
dx_group	Diagnosis group of diag_1
A1C_result	HbA1c category: '>8', '>7', 'Norm', or 'None' (test not performed)
days_to_next_encounter	Days until the patient's next recorded encounter in the health system
readmitted	Readmission status: '<30' (within 30 days), '>30', 'NO'

A.5 labs.csv (long format)

Column	Description
lab_id	Result identifier

Column	Description
admission_id	Admission the sample belongs to
test_name	Test performed
result_value	Reported result
unit	Unit of the reported result
collected_at	Collection timestamp

A test may be repeated during one stay. Eight tests are reported, with the usual reporting unit and an approximate adult reference interval (for orientation only; plausibility limits are yours to define):

Test	Usual unit	Approximate reference interval
Glucose (serum)	mg/dL	70–140
HbA1c	%	below 5.7 normal; 6.5 or more indicates diabetes
Creatinine	mg/dL	0.6–1.3
Sodium	mmol/L	135–145
Potassium	mmol/L	3.5–5.0
Hemoglobin	g/dL	12–17.5
Blood urea nitrogen (BUN)	mg/dL	7–20
White blood cell count (WBC)	10^3 cells/µL	4–11

A.6 medications.json

A list of records, one per admission:

- admission_id: admission identifier.
- prescriptions: list of orders, each with drug_name, dose, frequency, route and dose_change (Steady, Up, Down, New).
- insulin: nested object with regimen and change, or null if no insulin.
- allergies: documented allergies.
- discharge_notes: free text written by clinical staff.

A.7 Code lists

admission_type_id	Meaning
1 / 2 / 3	Emergency / Urgent / Elective
5 / 6 / 8	Not available / NULL / Not mapped
7	Trauma centre

discharge_disposition_id	Meaning

discharge_disposition_id	Meaning
1	Discharged to home
2	Transferred to another short-term hospital
3	Skilled nursing facility
4	Intermediate care facility
5	Other inpatient care
6	Home with home-health service
11	Expired
13	Hospice / home
18 / 25	NULL / Not mapped
22	Rehabilitation facility

admission_source_id	Meaning
1 / 2 / 3	Physician / clinic / HMO referral
4 / 5 / 6	Transfer from hospital / skilled nursing facility / other facility
7	Emergency room
17	NULL

payer_code: MC Medicare, MD Medicaid, HM HMO, BC Blue Cross, SP self-pay, CP commercial plan, OG other government, UN unknown.

ICD-9-CM chapters by 3-digit range (the data contain a subset of codes): 001–139 infectious; 140–239 neoplasms; 240–279 endocrine and metabolic (250 = diabetes mellitus); 280–289 blood; 290–319 mental; 320–389 nervous system and sense organs; 390–459 circulatory; 460–519 respiratory; 520–579 digestive; 580–629 genitourinary; 630–679 pregnancy; 680–709 skin; 710–739 musculoskeletal; 740–759 congenital; 760–779 perinatal; 780–799 symptoms and ill-defined conditions; 800–999 injury and poisoning. Codes starting with V are supplementary factors; codes starting with E are external causes.

A.8 The target

readmitted has three values. For this assignment readmitted_30d = 1 for '<30' and 0 otherwise. Roughly one stay in nine is a 30-day readmission, so the positive class is the minority.

A.9 Things the dataset does not give you

No cleaned reference version, no answer key, and no list of known problems. Finding, quantifying and handling the problems is the assignment.

Appendix B: Decision-log template (one row per decision)

Stage	Problem (with evidence)	Options considered	Choice	Why	Measured impact	Risk or limitation
e.g. 2						